



Midterm 161122 statmeth solutions

Statistical methods for AI (Vrije Universiteit Amsterdam)



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Midterm exam Statistical Methods – Solutions

22 November 2016

1.

- a) Incorrect. About 95% of the data points in a sample originating from a normal distribution will fall within 2 standard deviations of the population mean.
- b) Incorrect. The sample mean cannot be computed for ordinal data.
- c) Correct. In the formula for sample standard deviation we divide by $n - 1$, so for $n = 1$ the sample standard deviation cannot be computed.
- d) Correct. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ and $P(A \cap B)$ is nonnegative, so $P(A \cup B) \leq P(A) + P(B)$.

NB. Other examples/arguments are of course also possible.

2.

- a) In the table of measurements, 13 of them exceed 30 seconds. Therefore, an estimate based on the relative frequency is $\frac{13}{25} = 0.52$.
 - b) The amount of time spent on the main page is assumed to be normally distributed with mean $\mu = 30.00$ and standard deviation $\sigma = 10.00$. Therefore, the z -score of 28 seconds is $z = \frac{28.00 - 30.00}{10.00} = -0.20$. We use Table 2 and see that $P(Z \leq -0.20) = 0.4207$, where Z denotes the standard normal distribution. Hence, the probability that a single randomly selected visitor will spend less than 28 seconds on the main page of the portal is 0.4207 (or 42.07%).
 - c) The mean amount of time spent on the main page by $n = 36$ randomly selected visitors is normally distributed with mean $\mu = 30.00$ and standard deviation $\sigma = 10.00/\sqrt{36} = \frac{5}{3}$, by the Central Limit Theorem. Therefore, the z -score of 28 seconds is $z = \frac{28.00 - 30.00}{5/3} = -1.20$. We use Table 2 again and see that $P(Z \leq -1.20) = 0.1151$. Hence, the probability that the mean time spent on the main page by 36 randomly selected visitors is less than 28 seconds is 0.1151 (or 11.51%).
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3.

- a) Recall that $A = \{\text{Verge crashed at least once}\}$ and $B = \{\text{Marble is installed}\}$. We have $P(A) = 0.80$, and $P(\bar{A}) = 1 - P(A) = 0.20$, where $\bar{A}$ is the complement of the event A . We also have $P(B|A) = 0.92$ and $P(B|\bar{A}) = 0.46$. By the law of total probability

$$P(B) = P(B|A) \cdot P(A) + P(B|\bar{A}) \cdot P(\bar{A}) = 0.92 \cdot 0.80 + 0.46 \cdot 0.20 = 0.828.$$

- b) We are interested in $P(A|B)$. By Bayes' theorem

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B|A) \cdot P(A) + P(B|\bar{A}) \cdot P(\bar{A})}.$$

Therefore

$$P(A|B) = \frac{0.92 \cdot 0.80}{0.92 \cdot 0.80 + 0.46 \cdot 0.20} = \frac{2 \cdot 4 \cdot 0.46 \cdot 0.20}{2 \cdot 4 \cdot 0.46 \cdot 0.20 + 0.46 \cdot 0.20} = \frac{8}{9} \approx 0.889.$$

- c) Crashes of Verge increase the probability of installing Marbles, since $P(B|A) = 0.920 > 0.828 = P(B)$. It also implies that A and B are dependent, because if they were independent, $P(B|A) = P(B)$.

4.

- a) $\Omega = \{(1, H), (2, H), (3, H), (4, H), (5, H), (6, H), (1, T), (2, T), (3, T), (4, T), (5, T), (6, T)\}$. Both the die and the coin are fair, so all outcomes are equally likely and $P(\omega) = 1/12$ for all ω in Ω .
- b) The roll and the toss are independent by definition, so $P((i, H)) = \frac{1}{6} \cdot P(H)$ and $P((i, T)) = \frac{1}{6} \cdot P(T)$, where $i = 1, 2, \dots, 6$. Therefore all six outcomes with *Heads* have probability $\frac{1}{6} \cdot 0.6 = \frac{1}{10}$ and all six outcomes with *Tails* have probability $\frac{1}{6} \cdot 0.4 = \frac{1}{15}$.
- c) We find what Bridget earns for each outcome, and get the following table:

x	$\{X = x\}$	$P(\{X = x\})$
3	$\{(1, H), (2, H), (3, H), (4, H), (5, H), (6, H)\}$	$6 \cdot 1/12 = 1/2$
-1	$\{(1, T)\}$	$1/12$
-2	$\{(2, T)\}$	$1/12$
-3	$\{(3, T)\}$	$1/12$
-4	$\{(4, T)\}$	$1/12$
-5	$\{(5, T)\}$	$1/12$
-6	$\{(6, T)\}$	$1/12$

- d) By definition of expectation we get

$$E(X) = \sum_x x \cdot P(X = x) = 3 \cdot \frac{1}{2} - 1 \cdot \frac{1}{12} - 2 \cdot \frac{1}{12} - 3 \cdot \frac{1}{12} - 4 \cdot \frac{1}{12} - 5 \cdot \frac{1}{12} - 6 \cdot \frac{1}{12} = -0.25.$$

5.

- a) The histogram of Sample A resembles the bell-shaped curve. On the other hand, the other histogram indicates that Sample B comes from a population distribution that concentrates a lot of mass in the center (and looks quite uniform). Therefore, both tails of the population distribution of Sample A will be heavier than both tails of the population distribution of Sample B .
- b) The bottom line is the first quartile, the middle line is the second quartile (in other words, the median), and the top line is the third quartile.
- c) Sample A – normal QQ-plot 2, Sample B – normal QQ-plot 1. Both samples seem quite symmetric, and this automatically rules out the third normal QQ-plot. Sample A seems quite normal, and Sample B has lighter tails than normal distribution.